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Introduction to Nutritional Characterization of Proteins

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1. Introduction to Key Concepts for the Project

Before diving into the project details, it's important to understand key terms related to protein analysis and regulation. The following definitions will help clarify the concepts that are central to this work.

1.1. Amino Acids (AAs)

- **What are Amino Acids?**

Amino acids are organic molecules that form the building blocks of proteins, essential for various biological processes in all living organisms. The human body requires 20 amino acids (Appendix A) to function properly, and these can be grouped into indispensable (essential), dispensable (non-essential), and conditionally indispensable (essential) amino acids.

- **Essential Amino Acids:**

There are 9 essential amino acids that the body cannot produce on its own, so they must be obtained through dietary proteins. These include histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, and valine. Foods rich in essential amino acids include animal-based products like beef, eggs, and dairy, as well as plant-based sources like beans and nuts.

- **Non-Essential Amino Acids:**

The body can synthesize these amino acids, even if they are not directly consumed through food. Examples include alanine, arginine, cysteine, glutamic acid, glycine, and serine.

- **Conditionally Essential Amino Acids:**

These amino acids are generally non-essential but can become essential during growth and development, as well as periods of illness or stress. Examples include arginine, cysteine, glutamine, and tyrosine.

- **Role of Amino Acids in the Body:**

Amino acids are vital for the synthesis of proteins that perform key roles such as tissue growth, repair, and maintenance, hormone production, food break down, immune system support, and energy provision. They are also involved in processes like neurotransmitter production and digestion. Much like letters in the alphabet combine to form different words, amino acids combine in different sequences to create thousands of unique proteins, each with specific functions in the body.

1.2. Amino Acid Score (AAS)

- **What is AAS?**

The Amino Acid Score (AAS) is a measure used to evaluate the quality of a protein based on its essential amino acid content. The AAS compares the amount of each essential amino acid in a food protein to a reference protein provided by the Food and Agricultural Organization of the WHO. The score helps to determine how well a food protein can meet the body's amino acid requirements.

- **Why AAS is Important:**

A protein's AAS indicates whether it contains all the essential amino acids in the right proportions. If a protein has an AAS of 1, or 100%, it is considered a high-quality protein, meaning it contains all the necessary amino acids in amounts that meet the body's needs.

- **How AAS is Calculated:**

$$\text{AAS} = (\text{mg of limiting amino acid in 1 g of test protein} / \text{mg of the same amino acid in 1 g of reference protein}) \times 100.$$
 The lowest AAS value reflects the "**limiting**" amino acid, which is the essential amino acid present in the smallest proportion compared to the reference protein. The limiting amino acid determines the overall protein quality.

1.3. Protein Digestibility Corrected Amino Acid Score (PDCAAS)

- **What is PDCAAS?**

PDCAAS is a method that builds upon the AAS by incorporating the digestibility of a protein, offering a more complete picture of its nutritional value. Digestibility refers to how well a protein is broken down and absorbed by the body after consumption and is measured using an animal model (Human, pig, chicken, rat).

- **Why Digestibility is Important:**

The digestibility of a protein influences how much of the amino acids the body can actually use. Some proteins, particularly from animal sources, are more easily digested, while plant-based proteins may have lower digestibility due to factors like fiber content, and the presence of chemicals that inhibit digestive enzymes.

- **How PDCAAS is Calculated:**

$PDCAAS = (AAS \times \text{Digestibility})$. For example, if a protein has an AAS of 80% and a digestibility of 90%, its PDCAAS would be 72%. This means that after adjusting for digestibility, the protein supplies 72% of the essential amino acids needed by the body. PDCAAS values range from 0 to 1.0, with 1.0 representing a protein that provides 100% or more of the essential amino acids after digestion.

- **Applications of PDCAAS:**

PDCAAS was developed by the Food and Agriculture Organization (FAO) and World Health Organization (WHO) and is widely used to determine the protein quality of food products. This method is the standard adopted by North America for assessing human protein needs and regulating protein content claims.

1.4. In Vitro Protein Digestibility Corrected Amino Acid Score (IV-PDCAAS)

- **What is IV-PDCAAS?**

IV-PDCAAS refers to the **In Vitro** (outside of a living organism) method of calculating the Protein Digestibility Corrected Amino Acid Score. This lab-based approach simulates the digestive process to estimate protein digestibility without conducting direct animal studies.

- **How IV-PDCAAS Works:**

In vitro techniques mimic the action of digestive enzymes and conditions found in the human digestive system. After estimating digestibility, the IV-PDCAAS is calculated in the same way as with traditional methods ($\text{AAS} \times \text{in vitro digestibility}$). IV-PDCAAS is a quicker, more cost-effective way to assess protein quality, especially in the early stages of food product development.

- **Limitations of IV-PDCAAS:**

While IV-PDCAAS provides useful insights, it may not fully capture the complexities of real digestion in humans. Therefore, it is often used as a screening tool before conducting more detailed studies.

1.5. Labelling Regulations in North America

- **Labelling Terms for Protein Quality:**

In North America, food products may be labelled as a "source," "good source," or "excellent source" of protein, depending on the protein content and quality in relation to regulatory guidelines. These designations are based on both the amino acid profile and the digestibility of the protein (as assessed through methods like PDCAAS).

- **Regulatory Requirements:**

To determine if a protein source meets these labelling criteria, calculations from the PDCAAS score are used in conjunction with the Reference Amount Customarily Consumed (RACC), or Reference Amount in Canada. The RACC is the standardized serving size for different food categories and can be found in databases provided by regulatory agencies like Health Canada and the U.S. Food and Drug Administration (FDA). Meeting these guidelines allows manufacturers to make specific protein claims on food packaging, which can influence consumer choices.

1.6. Understanding DOIs and Their Use in Citations

- **What is a DOI?**

A DOI (Digital Object Identifier) is a unique string of numbers, letters, and symbols that permanently identifies an article or document.

- **Why Use a DOI?**

Unlike URLs, which may change, a DOI remains constant, ensuring reliable access to the specific source over time.

- **Where to Find a DOI?**

In scholarly articles, the DOI is typically printed on the first page, either below the title or in the header/footer. It can appear as:

- A standalone identifier starting with "10"
- A URL format starting with "<https://doi.org/>" or "<http://doi.org/>" followed by the DOI number.

- **How to Use a DOI in Citations?**

Incorporating a DOI in citations guarantees that the source can always be located, even if its web address changes.

2. Project Definition Overview

This project is structured around four key phases, each designed to progressively build toward the successful detection and analysis of proteins. The following outlines the purpose and focus of each phase.

Phase 1. Amino Acid Data Extraction:

Extract manuscript title, its **doi** if available, protein content (CP), amino acid content data from tables provided in published scientific papers. Provide a spreadsheet containing the extracted information. A template for the expected output is provided in Appendix B, Table B1.

Phase 2. Amino Acid Score (AAS) Calculation:

Calculate the Amino Acid Score (AAS) using different reference patterns, Appendix C. Summarize the results in a table or graph, highlighting the limiting amino acid (the one with the lowest score) for each reference pattern. A template for the expected output is provided in Appendix B, Table B2. If you choose to use visualization methods, please create a separate plot for each reference pattern.

Phase 3. Protein Digestibility Data Extraction:

Extract protein digestibility data, similar approach as step 1. Provide a spreadsheet that includes the digestibility of the protein source and the method used to measure it (in vitro or in vivo). Combine this digestibility data with the AAS from step 2 to calculate the Protein Digestibility-Corrected Amino Acid Score (PDCAAS) or In Vitro PDCAAS (IV-PDCAAS), whichever applies. Use Table B2 and add TPD, IVPD, PDCAAS, or IVPDCAAS (whichever applies) to the right of the DOI column. For teams working exclusively on this phase of the project, omit reference patterns from Template B2, as they do not apply.

Phase 4. Regulatory Compliance for Protein Labelling:

Assess whether the protein source meets the regulatory requirements for labelling as a "good

source," or "excellent source" of protein in North America. This assessment will be based on the PDCAAS or IV-PDCAAS score, along with serving size data from reference databases of [Health Canada](#) and compared against the regulatory requirements to meet “good source” or “excellent source.” This threshold for “good source” is 5-9.9 g of corrected protein content, while ≥ 10 meets the requirements for “excellent source.” Hint: Use the weight of the dry legumes and flours, not the processed weight (e.g., canned or baked). Please remove both "Model" columns from the Excel sheet provided to you.

3. Sample Hand Calculation

This section shows a step-by-step example of how to perform calculations related to this project. Each step is explained to help you understand the process involved in protein analysis.

Step 1: Extract data from paper

Paper Title: Effect of processing on the in vitro and in vivo protein quality of red and green lentils (Lens culinaris)

DOI: <https://doi.org/10.1016/j.foodchem.2017.07.129>

Tables: Tables 1 and 2 are adapted from Tables 1 and 3 of the manuscript. The latter includes all types of amino acids and protein (%CP), while Table 2 presents AAS, %TPD, and IVPD.

Table 1: Table of all types of amino acids and protein

	%DM ^a	%CF ^c	%CP ^b	ASP	THR	SER	GLU	PRO	GLY	ALA	CYS	VAL	MET	ILE	LEU	TYR	PHE	HIS	LYS	ARG	TRP
Casein	93.56	0.20	86.48	7.78	3.35	5.64	20.05	9.77	1.35	3.16	0.78	5.02	1.45	3.84	8.39	4.83	4.59	2.74	6.96	3.12	1.08
<i>Red Lentil</i>																					
Untreated	92.12	1.78	25.13	2.69	0.78	1.21	3.55	0.60	0.80	0.98	0.22	0.97	0.206	0.77	1.67	0.63	1.09	0.60	1.43	1.88	0.20
Extruded	95.41	1.08	26.86	3.30	0.96	1.47	4.38	0.79	0.97	1.25	0.24	1.18	0.22	0.96	1.88	0.68	1.33	0.79	1.81	2.01	0.20
Cooked	99.57	1.62	26.62	3.31	0.96	1.56	4.41	0.91	0.85	1.23	0.22	1.21	0.21	1.03	2.19	0.71	1.43	0.77	1.83	2.28	0.22
Baked	97.49	2.34	25.93	3.25	1.00	1.54	4.47	0.96	1.00	1.29	0.20	1.13	0.19	1.00	2.05	0.67	1.26	0.80	1.57	2.30	0.20
<i>Green Lentil</i>																					
Untreated	91.38	1.13	23.93	2.81	0.83	1.31	3.86	0.90	0.87	1.15	0.20	1.02	0.19	0.89	1.75	0.67	1.17	0.66	1.61	2.22	0.18
Extruded	95.13	1.48	24.65	3.05	0.89	1.36	4.02	0.73	0.91	1.20	0.21	0.99	0.20	1.02	1.93	0.64	1.24	0.73	1.71	2.14	0.21
Cooked	99.47	2.06	25.67	3.07	0.91	1.45	4.00	0.82	0.90	1.20	0.20	1.19	0.20	1.00	2.05	0.67	1.32	0.71	1.76	2.11	0.21
Baked	97.32	2.21	25.44	2.53	0.79	1.24	3.67	0.53	0.88	1.14	0.18	1.13	0.18	0.89	1.69	0.57	1.06	0.67	1.37	1.88	0.20

Table 2: Table of amino acid score, in vivo and in vitro protein

	Adj. Per	AAS ^a	%TPD ^b	IVPD ^c	PDCAAS ^d	In Vitro PDCAAS ^e
Casein	2.50	1.03	96.11 (1.4)	91.36 (0.7)	99.09	94.19
<i>Red Lentil</i>						
Extruded	1.05	0.68	92.38 (2.3)	88.01 (2.8)	63.01	60.03
Cooked	1.14	0.63	90.95 (2.2)	84.67 (1.1)	57.40	53.43
Baked	0.79	0.61	88.80 (2.4)	85.03 (0.5)	53.84	51.55
<i>Green Lentil</i>						
Extruded	1.08	0.66	86.02 (2.0)	84.30 (0.0)	57.09	55.95
Cooked	0.98	0.61	86.42 (2.0)	84.03 (1.2)	52.92	51.46
Baked	0.88	0.57	83.05 (1.9)	79.33 (1.0)	47.14	45.03

Step 2: Amino Acid Data Preparation

- **Table Notes:** Met (Methionine) and Cys (Cysteine) are linked chemically, as are Phe (Phenylalanine) and Tyr (Tyrosine). These pairs are treated as one source when calculating nutritional quality. Create columns for M+C (Met + Cys) and P+T (Phe + Tyr) by summing the values.
- If any amino acids are not reported, leave the corresponding cell blank.

Table 3: Example Data for Untreated Red Lentil (Essential Amino Acids, g/100g)

Sample	Protein %	THR	VAL	M+C	ILE	LEU	P+T	HIS	LYS	TRP	TPD%	IVPD
Red Lentil Untreated	25.13	0.78	0.97	0.426	0.77	1.67	1.72	0.60	1.43	0.20	-	-
Red Lentil Extruded	26.86	0.96	1.18	0.46	0.96	1.88	2.01	0.79	1.81	0.20	92.38	88.01

Step 3: Calculate Essential Amino Acids (mg/g protein)

- Use the formula:

$$\text{Essential Amino Acid (g/100g)} \times 1000 / \text{Protein (\%)}$$
- Example for THR (Threonine):

$$0.78 \times 1000 / 25.13 = 31.04$$

Table 4 lists the resulting essential amino acids in mg/g of protein.

Table 4: Essential Amino Acids of untreated and extruded red lentil in mg/g protein

Sample	THR	VAL	M+C	ILE	LEU	P+T	HIS	LYS	TRP
Red Lentil Untreated	31.04	38.60	16.95	30.64	66.45	68.44	23.88	56.90	7.96
Red Lentil Extruded	35.74	43.93	17.13	35.74	70.0	74.83	29.41	67.39	7.45

Step 4: Compare to reference pattern.

Table 5 displays the FAO 91 Reference Pattern for ages 10-12 years (see Appendix B).

Table 5: FAO 91 Reference Pattern for 10-12 yrs.

THR	VAL	M+C	ILE	LEU	P+T	HIS	LYS	TRP
28	25	22	28	44	22	19	44	9

- For each amino acid, calculate:
(mg of amino acid in test protein) / (mg of amino acid in reference protein)
- Example for THR:

$$31.04 / 28 = 1.1085$$

The resulting Amino Acid Scores (AAS) are presented in Table 6.

- The **limiting amino acid** is the one with the lowest AAS.

Table 6: AAS for Untreated and extruded red lentil for the age of 10-12 years.

Sample	THR	VAL	M+C	ILE	LEU	P+T	HIS	LYS	TRP	AAS
Red Lentil Untreated	1.11	1.54	0.77	1.09	1.51	3.11	1.26	1.29	0.88	0.77
Red Lentil Extruded	1.28	1.76	0.78	1.28	1.59	3.40	1.55	1.53	0.83	0.78

Step 5: Calculate PDCAAS (Protein Digestibility Corrected Amino Acid Score)

Lentils that haven't been heat-treated are inedible, which is why there is no digestibility value for "Untreated Red Lentil." As a result, we can't perform an in vivo digestibility analysis on them.

For the Extruded Red Lentil, in-vivo and in-vitro digestibility values are TPD = 92.38 % and IVPD = 88.01%, respectively. The AA Score is 0.78. Then

$$\text{PDCAAS} = \text{AAS} \times \text{TPD} = 0.78 \times 92.38 = 72.06$$

$$\text{IV-PDCASS} = \text{AAS} \times \text{IVPD} = 0.78 \times 88.01 = 68.64$$

This calculation shows that the in vivo and in vitro protein quality of extruded red lentils, when corrected for digestibility, are 72.06% and 68.64%, respectively.

Step 6: Protein Content Claim Calculation

This calculation indicates the amount of "quality" protein present in a food item based on the amount typically consumed. There are two thresholds for protein content claims using PDCAAS. If the PDCAAS-corrected protein per Reference Amount Customarily Consumed (RACC) is between 5 and 9.9 grams, the food is considered a **Good Source of Protein**. If it is 10.0 grams or more, it is considered an **Excellent Source of Protein**.

How is this calculated?

Three factors are required:

1. **Protein Content** – This can be expressed as a percentage (g/100g sample) in the provided manuscripts. If expressed as a percentage, it should be converted before being used in the calculation.
2. **PDCAAS** – This can also be expressed as a percentage in the provided manuscripts and should be converted before being used in the calculation.
3. **RACC** – This is provided by Health Canada on a gram basis.

Formula: Content Claim Suitability = Protein Content × PDCAAS × RACC

Examples :

Tables 7 and 8 display the protein content claim calculations for almonds and chickpeas, respectively.

Calculation for Almonds: Using the above values, correcting for protein content as a percentage and PDCAAS as a percentage:

Content Claim Suitability = Protein Content $\times$ PDCAAS $\times$ RACC = $(21/100) \times (39/100) \times 30 = 2.457 \approx 2.5\text{g}$

Since 2.5 g is below the 5 g threshold, this product does not qualify for a protein content claim.

Calculation for Chickpeas: Using the above values, correcting for protein content as a percentage and PDCAAS as a percentage:

Content Claim Suitability = Protein Content $\times$ PDCAAS $\times$ RACC = $(23/100) \times (70/100) \times 35 = 5.957 \approx 6.0\text{g}$

Since 6.0 g exceeds the 5 g threshold, this product qualifies for a protein content claim and is considered a Good Source of Protein.

Table 7: Protein Content Claim Calculation for Almonds

Protein Source	Protein Content (%)	PDCAAS (%)	RACC (g)	Corrected Protein Content in RACC (g)	Content Claim Status
Almonds	21	39	30	2.5	No Claim

Table 8: Protein Content Claim Calculation for Chickpeas

Protein Source	Protein Content (%)	PDCAAS (%)	RACC (g)	Corrected Protein Content in RACC (g)	Content Claim Status
Chickpeas	23	70	35	6.0	Good Source

References

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Appendix A: Amino Acids

Table A.1 Amino acids, one and three letter codes

Amino acid	Three letter symbol	One letter symbol*
Alanine	Ala	A
Arginine	Arg	R
Asparagine	Asn	N
Aspartic acid	Asp	D
Cysteine	Cys	C
Glutamic acid	Glu	E
Glutamine	Gln	Q
Glycine	Gly	G
Histidine	His	H
Isoleucine	Ile	I
Leucine	Leu	L
Lysine	Lys	K
Methionine	Met	M
Phenylalanine	Phe	F
Proline	Pro	P
Serine	Ser	S
Threonine	Thr	T
Tryptophan	Trp	W
Tyrosine	Tyr	Y
Valine	Val	V

Appendix B: Templates

Project 1

Instruction: Extract information from the paper and create a table containing the following details:

- List amino acids using their 3-letter abbreviations.
- According to the referenced paper, arrange the sample name as either {process} {bean} or {bean} {process} (e.g., *Baked Green Lentil* or *Green Lentil Baked*). Ensure you remain consistent with the way samples are presented in the reference paper.

Table B1. Phase 1 output template

Sample Name	Protein (%)	ASP	...	TRP	DOI
Baked Green Lentil	25.44	2.53	...	0.20	DOI link

Table B2. Phase 2 output template

Sample Name	THR	VAL	M + C	...	TRP	AAS	Limiting Amino Acid	Reference Pattern 1991			Reference Pattern 2007			Reference Pattern 2013			doi
								Infant	...	Adult	0.5	...	>18yr	Infant (0-6mo)	...	>3yr	
Red Lentil Untreated	1.11	1.54	0.77	1.09	0.88	0.77	M + C										Doi link

Appendix C: Amino Acids Reference Patterns

All values in the tables are expressed as mg of protein per gram.

Table C1. FAO Food and Nutrition Paper 51 (Table 7) – year 1991

	Infant	Pre-School Child 2-5 yrs	School –Child 10-12 yrs	Adult
Histidine	26	19	19	16
Isoleucine	46	28	28	13
Leucine	93	66	44	19
Lysine	66	58	44	16
Met+Cys	42	25	22	17
Phe+Tyr	72	63	22	19
Threonine	43	34	28	9
Tryptophan	17	11	9	5
Valine	55	35	25	13

Table C2. WHO TRS 935 (Table 36, Modified in Protein Quality report 92 Table 3) – year 2007

	0.50	1-2 yr	3-10 yr	11-14 yr	15-18 yr	>18yr
Histidine	20	18	16	16	16	15
Isoleucine	32	31	31	30	30	30
Leucine	66	63	61	60	60	59
Lysine	57	52	48	48	47	45
Met+Cys	27	25	23	23	23	22
Phe+Tyr	52	46	41	41	40	38
Threonine	31	27	25	25	24	23
Tryptophan	8.5	7.4	6.6	6.5	6.3	6
Valine	43	42	40	40	40	39

Table C3. Protein Quality Report 92 (Table 5) – year 2013

	Infant (0-6 mo)	Child (6mo-3 yr)	> 3 yr
Histidine	21	20	16
Isoleucine	55	32	30
Leucine	96	66	61
Lysine	69	57	48
Met+Cys	33	27	23
Phe+Tyr	94	52	41
Threonine	44	31	28
Tryptophan	17	8.5	6.6
Valine	55	43	40